Volatility, Correlation Regimes and Value-at-Risk in Saudi Equity Sectors, 2015–2026

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Abstract

This paper applies standard financial-engineering risk tools to Saudi Exchange (Tadawul) equities. Using split- and dividend-adjusted daily prices for 20 large-cap constituents from 2015-01-04 to 2026-10-29 (2,978 trading sessions), I build six equal-weight sector baskets (Banks, Materials, Energy, Telecom, Food & Staples, Utilities) and an equal-weight large-cap composite. I estimate GARCH(1,1) and GJR-GARCH(1,1) models with Student-t innovations, track rolling correlations with the composite and with Brent crude, measure drawdowns, and compare historical, Gaussian and Student-t Value-at-Risk (VaR) and Expected Shortfall (ES). I then backtest one-day-ahead 95% and 99% VaR forecasts over 1,978 out-of-sample days using the Kupiec (1995) proportion-of-failures test and the Christoffersen (1998) independence test. Three results stand out. First, every series has fat tails (excess kurtosis between 4.7 and 9.0), and the leverage term in GJR-GARCH is significant at 5% for 5 of 7 series. Second, correlations jump in a crisis. The average pairwise sector correlation rose from 0.44 in 2016–19 to 0.77 in the COVID crash, and the weekly correlation between the composite and Brent rose from 0.22 to 0.63. Third, the Gaussian 99% VaR is rejected by the Kupiec test for 7 of 7 series, the rolling historical VaR for 5, and the GJR-GARCH-t VaR for 0. Volatility-aware, fat-tailed models are therefore needed for reliable tail-risk measurement in this market.

Keywords: Saudi Arabia, Tadawul, GARCH, GJR-GARCH, Value-at-Risk, Expected Shortfall, Kupiec test, correlation regimes, oil prices.

1. Introduction

The Saudi equity market is one of the largest in emerging markets. Its sectors are tied to oil in several ways: through petrochemical margins, through government spending that drives bank credit and consumption, and through investor sentiment. That makes it a useful setting for asking a practical risk question: how much do standard volatility and VaR models change by sector, and do they hold up when oil and global markets are under stress?

This paper is an independent, reproducible study. It does not offer a new estimator. It applies established methods (GARCH-family volatility models, rolling and regime-conditional correlations, drawdown analysis and VaR/ES with formal backtests) to a carefully documented public dataset. The code, data manifest, tables and figures are all published with the paper, and every number in this paper is generated from the pipeline output files. The goal is transparency rather than novelty.

I focus on four questions:

1. How do return distributions differ across Saudi sectors in mean, volatility, skewness and tail thickness?
2. Is volatility persistent and asymmetric (larger after negative shocks), and does that vary by sector?
3. How do cross-sector correlations, and correlation with Brent crude, change across oil and pandemic regimes?
4. Which VaR methods pass formal backtests at the 95% and 99% levels?

2. Data

2.1 Sources

All prices are public and come from Yahoo Finance through the open-source `yfinance` library (version pinned in `requirements.txt`), using `auto_adjust=True` so closes are adjusted for splits and dividends. The download timestamp, tickers and row count are recorded in `data/raw/MANIFEST.json` (download time 2026-10-30T12:54:01+00:00 UTC). The study uses:

- **20 Tadawul large caps** (`.SR` tickers) grouped into six sectors by their Saudi Exchange / GICS industry group:
- Banks: Al Rajhi (1120), Saudi National Bank (1180), Riyad Bank (1010), Alinma (1150), SAB (1060)
- Materials: SABIC (2010), Ma'aden (1211), SABIC Agri-Nutrients (2020), Saudi Kayan (2350)
- Energy: Saudi Aramco (2222, from its December 2019 listing), Petro Rabigh (2380), Bahri (4030), SARCO (2030)
- Telecom: stc (7010), Mobily (7020), Zain KSA (7030)
- Food & Staples: Almarai (2280), Savola (2050)
- Utilities: Saudi Electricity (5110), ACWA Power (2082, from its 2021 listing)
- **Brent crude**: ICE Brent front-month futures (`BZ=F`).
- **External benchmark**: iShares MSCI Saudi Arabia ETF (`KSA`, USD, NYSE Arca, from September 2015). The Saudi riyal is pegged to the US dollar, so currency effects are small.

The official TASI index (`^TASI.SR`) returned only one recent observation from Yahoo Finance when this study was built, and Stooq requires a browser check. I therefore do not use TASI directly. The large-cap equal-weight composite ("LargeCap EW") stands in for the market, and the KSA ETF is a cross-check.

2.2 Construction

Daily log returns are computed for each stock on its own trading days. Tadawul enforces a ±10% daily price limit, so any simple return larger than 10.5% in absolute value is a data or corporate-action artefact (for example rights issues, capital reductions or bad prints) rather than a tradable move. These 25 observations, out of roughly 59,560 stock-days, are set to missing and listed in `reports/tables/`

`flagged_observations.csv`. A session is kept when at least 10 of the 20 constituents report a price. That removes non-Tadawul calendar rows such as Fridays, which come from the US-listed and futures series.

Each sector basket is an equal-weight portfolio of its available constituents, rebalanced daily. Its log return is ln(1 + mean simple return of available members). Stocks join a basket on their listing date, so Energy includes Aramco only from December 2019 and Utilities includes ACWA Power only from 2021. The LargeCap EW composite is the same equal-weight average across all 20 names. For the oil analysis, weekly (Thursday-anchored) log returns are used because Tadawul (Sunday–Thursday) and Brent (Monday–Friday) trade on different calendars, which would bias daily correlations. The final sample has 2,978 daily and 608 weekly observations.

3. Methods

Return statistics. Annualised mean (250 sessions per year), volatility, Sharpe ratio with a zero risk-free rate, skewness, excess kurtosis, a Jarque–Bera normality test and the extreme daily moves.

Volatility models. For each series r_t (in percent), r_t = μ + ε_t with ε_t = σ_t z_t and z_t following a standardised Student-t with ν degrees of freedom. GARCH(1,1) (Bollerslev, 1986) sets σ²_t = ω + α ε²_{t-1} + β σ²_{t-1}. GJR-GARCH (Glosten, Jagannathan and Runkle, 1993) adds γ ε²_{t-1} 1[ε_{t-1}<0], so negative shocks raise variance by α+γ instead of α. Persistence is α+β+γ/2, and the half-life of a variance shock is ln(0.5)/ln(persistence). Models are estimated by maximum likelihood with the `arch` package (Sheppard), following its univariate volatility examples.

Correlations and regimes. I compute 126-day rolling correlations of each sector with the LargeCap EW composite and 52-week rolling correlations of weekly sector returns with Brent. I also compute correlations within six pre-defined regimes based on known oil and pandemic events rather than on the data: the 2015–16 oil slump (to February 2016), the 2016–19 recovery, the COVID crash (20 February to 30 June 2020, which includes the March 2020 OPEC+ price war), the 2020–21 post-COVID recovery, the 2022 oil spike, and 2023 to the present.

Drawdowns. Running peak-to-trough decline of cumulative wealth, with peak, trough and recovery dates.

VaR and ES. Full-sample one-day VaR and ES at 95% and 99% under three methods: historical (empirical quantile and tail mean), Gaussian (closed form), and Student-t fitted by maximum likelihood (closed-form ES). ES is the average loss beyond VaR, as in Acerbi and Tasche (2002) and the Basel FRTB framework.

VaR backtest. One-day-ahead forecasts use only data available up to t−1. (i) Historical: 250-day rolling empirical quantile. (ii) Normal: 250-day rolling mean and standard deviation. (iii) GJR-GARCH-t: re-estimated every 63 sessions on a rolling 1,000-day window, then filtered daily with fixed parameters between re-estimations.

The backtest covers the 1,978 days left after the 1,000-day burn-in. An exceedance occurs when the realised return falls below −VaR. The Kupiec (1995) POF likelihood ratio tests whether the exceedance rate equals the nominal rate ($\chi^2(1)$). The Christoffersen (1998) independence test checks whether exceedances cluster. All tests are at the 5% level.

4. Results

4.1 Return statistics

Series	N	Ann. return %	Ann. vol. %	Sharpe	Skew	Excess kurt.	Worst day %	Best day %
Banks	2978	10.09	19.66	0.49	-0.43	6.71	-9.39	8.11
Materials	2978	5.21	21.66	0.23	-0.25	5.01	-9.37	9.45
Energy	2978	7.27	22.56	0.31	-0.64	5.79	-8.66	8.63
Telecom	2978	6.10	20.08	0.29	-0.31	6.59	-8.37	9.01
Food & Staples	2978	1.31	23.36	0.06	0.17	4.67	-8.63	9.94
Utilities	2978	7.03	26.02	0.26	0.03	5.05	-10.04	9.93
LargeCap EW	2978	7.72	16.82	0.44	-0.79	9.02	-7.95	8.50
KSA ETF (USD)	2206	10.69	18.05	0.56	1.30	23.20	-9.31	16.02
Brent (weekly)	608	7.52	39.14	0.19	-0.88	6.47	-33.55	20.72

Table 1. Daily return statistics, 2015-01-04 to 2026-10-29. Brent statistics use weekly data annualised with 52 weeks.

Over the sample, Banks had the highest annualised return (10.1%) and Food & Staples the lowest (1.3%). Volatility ranges from 19.7% for Banks to 26.0% for Utilities. The LargeCap EW composite returned 7.7% a year with 16.8% volatility, which shows the diversification benefit of holding all sectors. Every series rejects normality (Jarque–Bera p-value below 0.001). Excess kurtosis is 9.0 for the composite and between 4.7 and 6.7 across sectors. Composite skewness is -0.79, so large losses happen more often than large gains of the same size. Weekly Brent volatility is 39.1% a year, well above any Saudi sector basket.

4.2 Volatility dynamics

Series	ω	α	γ	β	ν	p(γ)	Persist.	Half-life (d)	AIC GARCH	AIC GJR
Banks	0.098	0.043	0.127	0.837	4.309	0.000	0.943	11.862	8992.126	8966.671
Materials	0.076	0.048	0.094	0.870	4.681	0.000	0.965	19.628	9617.912	9597.332
Energy	0.081	0.102	0.095	0.824	4.539	0.003	0.973	25.703	9631.536	9620.248
Telecom	0.175	0.110	0.078	0.760	3.704	0.061	0.910	7.329	9022.956	9019.308
Food & Staples	0.166	0.076	0.102	0.811	4.188	0.001	0.938	10.819	10090.716	10080.318
Utilities	0.715	0.322	0.151	0.515	3.034	0.069	0.912	7.553	10543.989	10542.308
LargeCap EW	0.051	0.022	0.145	0.853	5.052	0.000	0.948	13.043	7869.809	7822.340

Table 2. GJR-GARCH(1,1)-t estimates (percent returns), with AIC compared against GARCH(1,1)-t.

Volatility is highly persistent. GJR persistence ranges from 0.910 to 0.973, which implies half-lives of 7.3 to 25.7 sessions. The asymmetry coefficient γ is positive for every series and significant at 5% for Banks, Materials, Energy, Food & Staples, LargeCap EW. GJR-GARCH has a lower AIC than symmetric GARCH for Banks, Materials, Energy, Telecom, Food & Staples, Utilities, LargeCap EW. In the

composite, γ = 0.145 while α = 0.022, so almost all of the news impact comes from negative shocks. The Student-t degrees of freedom range from 3.0 (Utilities has the fattest tails) to 5.1, which supports fat-tailed innovations even after volatility is accounted for.

Figure 2 shows conditional volatility. Every sector peaked during the COVID shock: Banks at 87.4% (2020-03-10), Energy at 82.5% (2020-03-11), Materials at 77.0% (2020-03-10), and the composite at 76.8% (2020-03-10), compared with a full-sample median of 13.8% for the composite. Most of these peaks fall around 9–11 March 2020, right after the OPEC+ talks collapsed on 6 March, which suggests the oil price war added to the pandemic shock in Saudi equities. At the end of the sample, conditional volatility is 25.3% for the composite and 40.7% for Utilities (median 24.9%).

4.3 Correlation regimes and oil

Regime	Days	Avg. pairwise corr.	Mkt ann. vol. %	Mkt cum. ret. %	Brent cum. ret. %	Mkt–Brent corr. (wkly)
Oil slump 2015-16	297	0.62	25.07	-23.66	-30.75	0.52
Recovery 2016-19	1015	0.44	14.86	74.60	57.26	0.22
COVID crash 2020	89	0.77	36.18	-3.36	-25.55	0.63
Post-COVID 2020-21	376	0.42	13.04	60.31	92.39	0.48
Oil spike 2022	248	0.45	17.71	3.35	0.61	0.44
2023-present	953	0.41	13.49	13.61	48.69	0.19

Table 3. Regime summary. Average pairwise sector correlation and market volatility use daily data; Brent figures use weekly returns.

Regime	Banks	Materials	Energy	Telecom	Food & Staples	Utilities
Oil slump 2015-16	0.47	0.43	0.53	0.36	0.48	0.34
Recovery 2016-19	0.17	0.30	0.16	0.09	0.16	0.10
COVID crash 2020	0.71	0.65	0.55	0.50	0.50	0.40
Post-COVID 2020-21	0.40	0.44	0.31	0.25	0.41	0.27
Oil spike 2022	0.43	0.39	0.35	0.23	0.19	0.29
2023-present	0.11	0.23	0.28	0.09	0.00	0.08

Table 4. Weekly correlation of each sector basket with Brent, by regime.

Cross-sector correlation depends on the regime. Average pairwise correlation was 0.62 during the 2015–16 oil slump, fell to 0.44 in the 2016–19 recovery, peaked at 0.77 in the COVID crash, and has been 0.41 since 2023. So diversification across Saudi sectors largely disappears in a crisis (Figure 8). The oil link follows a similar pattern. Weekly composite–Brent correlation was 0.52 in the oil slump, 0.22 in 2016–19, 0.63 in the COVID crash, 0.44 in 2022, and only 0.19 since 2023. During the COVID crash, Banks had the highest Brent correlation (0.71), above Energy (0.55). This fits a channel where oil drives fiscal spending, liquidity and credit, rather than a simple cash-flow link through energy companies. Since 2023, Brent correlations have been close to zero for Telecom (0.09), Food & Staples (0.00) and Utilities (0.08). The market returned 13.6% in that period, while Brent's cumulative weekly log-return sum translates to 48.7%.

4.4 Drawdowns

Series	Max DD %	Peak	Trough	Recovered	Current DD %
Banks	-44.1	2019-05-01	2020-03-16	2021-04-28	-14.0
Materials	-47.6	2015-05-04	2016-01-20	2018-01-15	-38.4
Energy	-56.3	2015-05-28	2016-10-18	2020-08-25	-7.4
Telecom	-39.4	2015-03-09	2016-01-17	2019-03-03	-14.5
Food & Staples	-49.5	2015-04-20	2016-01-21	2017-06-18	-29.1
Utilities	-48.6	2024-04-23	2026-03-01	not yet	-43.5
LargeCap EW	-43.7	2015-05-26	2016-01-21	2018-04-08	-10.9

Table 5. Maximum drawdowns, with recovery dates and drawdown at the end of the sample.

The deepest drawdown was in Energy (56.3%, from 2015-05-28 to 2016-10-18). The composite fell 43.7% from 2015-05-26 to 2016-01-21, during the 2015–16 oil slump, and recovered by 2018-04-08. That is a larger loss than the 2020 COVID drawdown. Banks had their largest drawdown in COVID (44.1%, trough 2020-03-16). Series that had not recovered their maximum drawdown by the end of the sample: Utilities. At the end of the sample, Utilities is 43.5% and Materials 38.4% below their previous peaks.

4.5 Static VaR and ES

Series	VaR Historical	VaR Normal	VaR Student-t	ES Historical	ES Normal	ES Student-t
Banks	3.41	2.85	3.45	5.06	3.27	5.18
Energy	4.53	3.29	3.99	6.25	3.77	6.10
Food & Staples	4.01	3.43	4.15	5.57	3.93	6.13
LargeCap EW	3.06	2.44	2.83	4.76	2.81	4.21
Materials	3.59	3.17	3.83	5.36	3.63	5.66
Telecom	3.56	2.93	3.62	5.22	3.36	5.68
Utilities	4.59	3.80	5.06	6.29	4.36	8.49

Table 6. Full-sample one-day 99% VaR and ES (% loss).

The Gaussian assumption understates tail risk. For the composite, the 99% Gaussian VaR is 2.44%, compared with a historical VaR of 3.06%. The gap is wider for ES: 2.81% under the Gaussian model against 4.76% historically and 4.21% under the Student-t. For every series, the Gaussian 99% ES is below the historical ES. Utilities carry the largest tail risk, with a historical 99% ES of 6.29%.

4.6 VaR backtest

Series	Historical rate %	Normal rate %	GJR-GARCH-t rate %	Historical Kupiec p	Normal Kupiec p	GJR-GARCH-t Kupiec p
Banks	1.37	1.92	1.01	0.12	0.00	0.96
Energy	1.67	1.97	1.06	0.01	0.00	0.78
Food & Staples	1.26	1.62	1.06	0.26	0.01	0.78
LargeCap EW	1.52	2.12	1.47	0.03	0.00	0.05
Materials	1.57	1.87	1.06	0.02	0.00	0.78
Telecom	1.67	2.12	0.71	0.01	0.00	0.17
Utilities	1.62	2.22	0.71	0.01	0.00	0.17

Table 7. One-day 99% VaR backtest over 1,978 out-of-sample sessions (expected rate 1%; Kupiec p-value).

At 99%, the Gaussian VaR is rejected by Kupiec for Banks, Materials, Energy, Telecom, Food & Staples, Utilities, LargeCap EW. Its exceedance rates run from 1.62% to 2.22%, 1.6× to 2.2× the 1% target. The rolling historical VaR is rejected for Materials, Energy, Telecom, Utilities, LargeCap EW. The GJR-GARCH-t VaR is rejected for none, with exceedance rates between 0.71% and 1.47%. Its weakest case is the composite (1.47%, $p = 0.051$), which is close to rejection. Exceedances cluster (Christoffersen $p < 0.05$) for Materials, Energy, LargeCap EW under historical VaR, Banks, Materials, Energy, Telecom, LargeCap EW under Gaussian VaR, and none under GJR-GARCH-t. Rolling-window methods are slow to react when volatility jumps, so their misses arrive in runs. At 95%, Kupiec rejects historical VaR for Materials, Gaussian VaR for Food & Staples, and GJR-GARCH-t for none, so model choice matters most deep in the tail. Figure 6 shows the GJR forecast widening quickly in March 2020, while the historical VaR adjusts late and then stays too conservative for a year.

5. Discussion

The results support three practical points. First, a 250-day Gaussian VaR, still common in simple dashboards, is not adequate for Saudi sector exposures at 99%. Second, conditional volatility with Student-t innovations and an asymmetric term fixes most of the coverage problem at little cost. Third, correlation-based diversification is weakest exactly when it is needed. Cross-sector and oil correlations both jumped during the 2015–16 and 2020 stresses, so stress tests should use crisis correlation matrices (Figure 8) rather than full-sample averages.

The weaker link between Brent and the market since 2023 is notable. It may reflect diversification of the economy and of the investor base, but this descriptive study cannot establish cause.

6. Limitations

- **Basket proxies, not official indices.** The sector baskets are equal-weight portfolios of 2 to 5 large caps, not the official Tadawul sector indices, which are cap-weighted and include many more stocks. Results for Food & Staples and Utilities (two names each) are especially concentrated. The official TASI series was not available from the free source used.
- **Survivorship and selection.** Constituents were chosen from today's large caps, which biases the sample toward firms that survived and grew. Aramco and ACWA Power enter part-way through the sample, which changes the Energy and Utilities baskets.
- **Data quality.** Yahoo Finance adjusted prices may contain errors. 25 observations were removed using the price-limit rule, and other errors may remain. Dividend adjustment follows Yahoo's method.
- **Calendar asynchrony.** Brent and the KSA ETF trade on different days from Tadawul. Weekly data reduces but does not remove this effect.

- **Model scope.** These are univariate models. Multivariate GARCH or DCC, extreme value theory and filtered historical simulation were not estimated. The backtests ignore parameter uncertainty and multiple-testing effects across 42 tests.
- **Regime dates** are set from known events, not estimated. Formal break tests (for example Bai–Perron or Markov switching) are left for future work.
- **No trading costs or liquidity effects.** The results describe risk. They are not a trading strategy or investment advice.

7. Conclusion

Using public data and open-source tools, this study shows that Saudi sector returns are fat-tailed, have persistent and mostly asymmetric volatility, and become much more correlated with each other and with oil in a crisis. Among the three VaR approaches tested, a GJR-GARCH model with Student-t innovations gives the most reliable 99% coverage across sectors. All code, tables and figures are in the accompanying repository and can be regenerated with one command.

References

- Acerbi, C. and Tasche, D. (2002). On the coherence of Expected Shortfall. Journal of Banking & Finance, 26(7), 1487–1503.
- Bollerslev, T. (1986). Generalized autoregressive conditional heteroskedasticity. Journal of Econometrics, 31(3), 307–327.
- Christoffersen, P. F. (1998). Evaluating interval forecasts. International Economic Review, 39(4), 841–862.
- Engle, R. F. (1982). Autoregressive conditional heteroscedasticity with estimates of the variance of United Kingdom inflation. Econometrica, 50(4), 987–1007.
- Glosten, L. R., Jagannathan, R. and Runkle, D. E. (1993). On the relation between the expected value and the volatility of the nominal excess return on stocks. Journal of Finance, 48(5), 1779–1801.
- Jorion, P. (2007). Value at Risk: The New Benchmark for Managing Financial Risk (3rd ed.). McGraw-Hill.
- Kupiec, P. H. (1995). Techniques for verifying the accuracy of risk measurement models. Journal of Derivatives, 3(2), 73–84.
- Basel Committee on Banking Supervision (2019). Minimum capital requirements for market risk (FRTB). Bank for International Settlements.
- Sheppard, K. arch: ARCH models in Python (software). https://github.com/bashtage/arch
- Data: Yahoo Finance via `yfinance` (Aroussi, R.), tickers listed in Section 2.1.

Figures

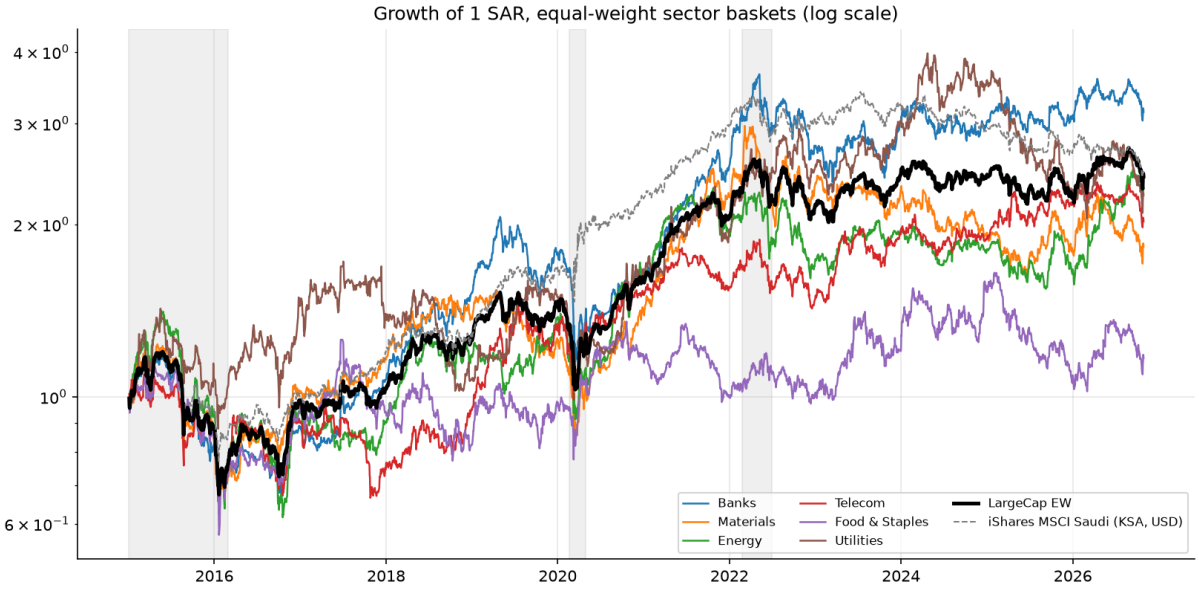


Figure 1. Cumulative growth of sector baskets and the LargeCap EW composite.

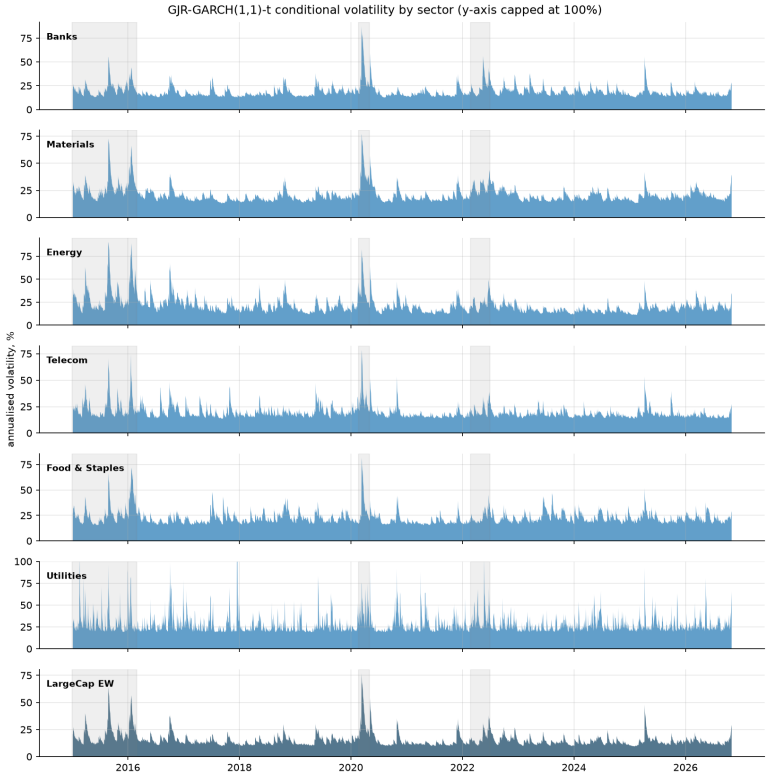


Figure 2. GJR-GARCH(1,1)-t annualised conditional volatility.

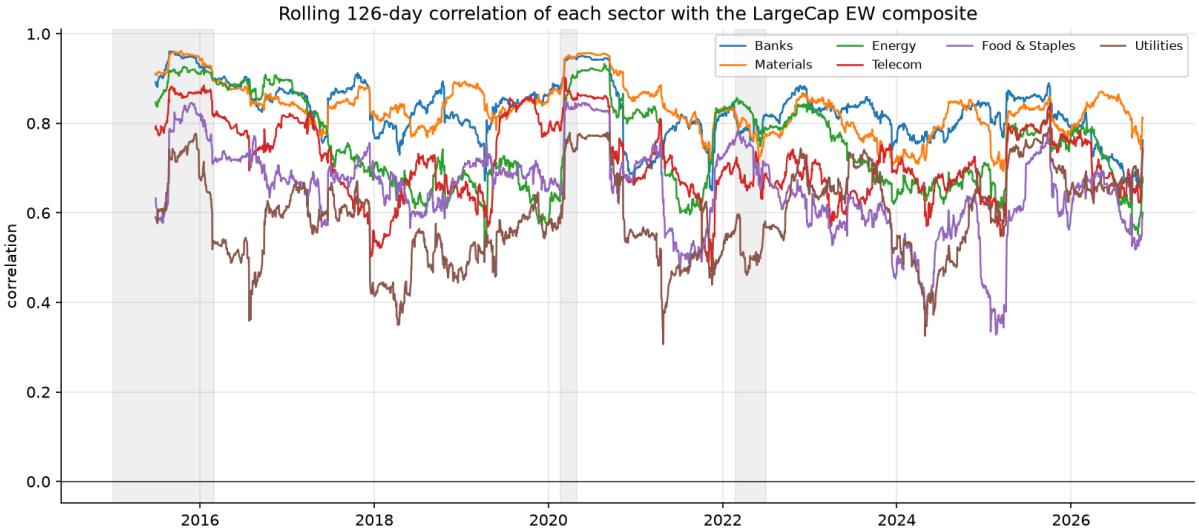


Figure 3. 126-day rolling correlation of each sector with the LargeCap EW composite.

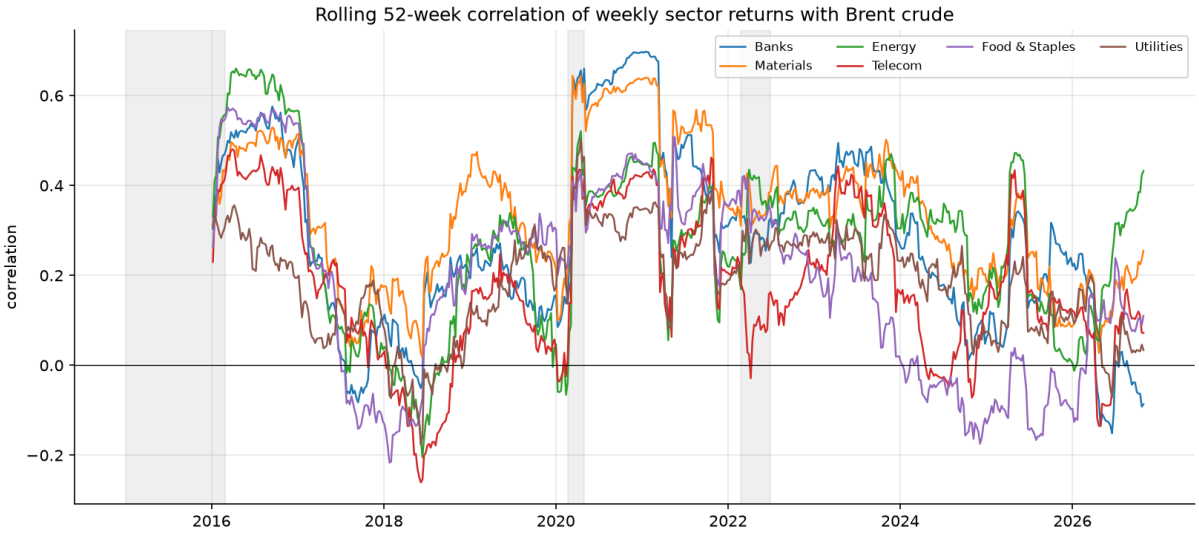


Figure 4. 52-week rolling correlation of weekly sector returns with Brent.

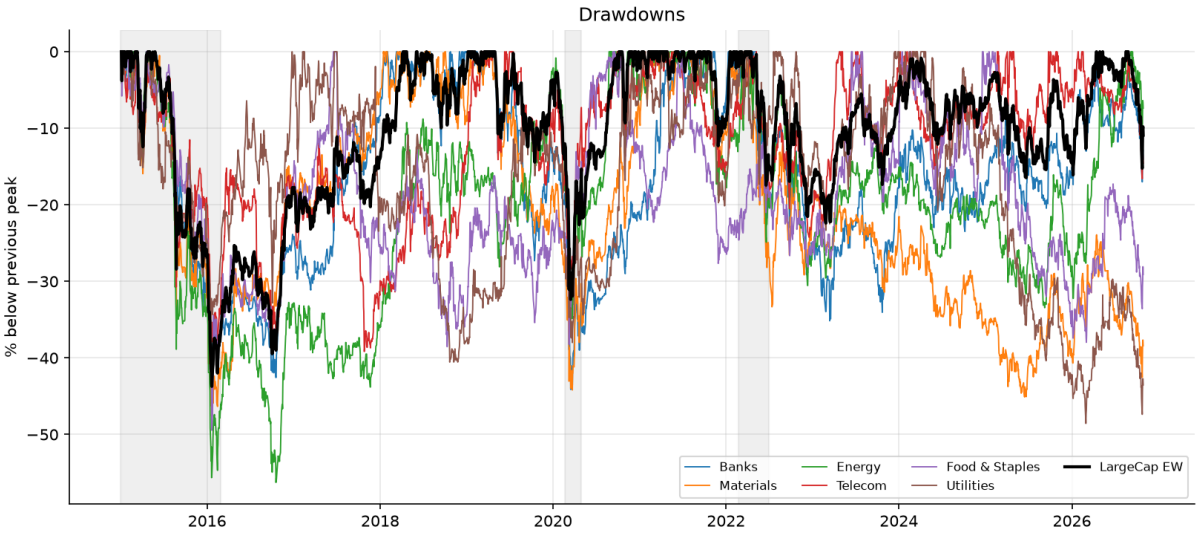


Figure 5. Drawdowns from running peak.

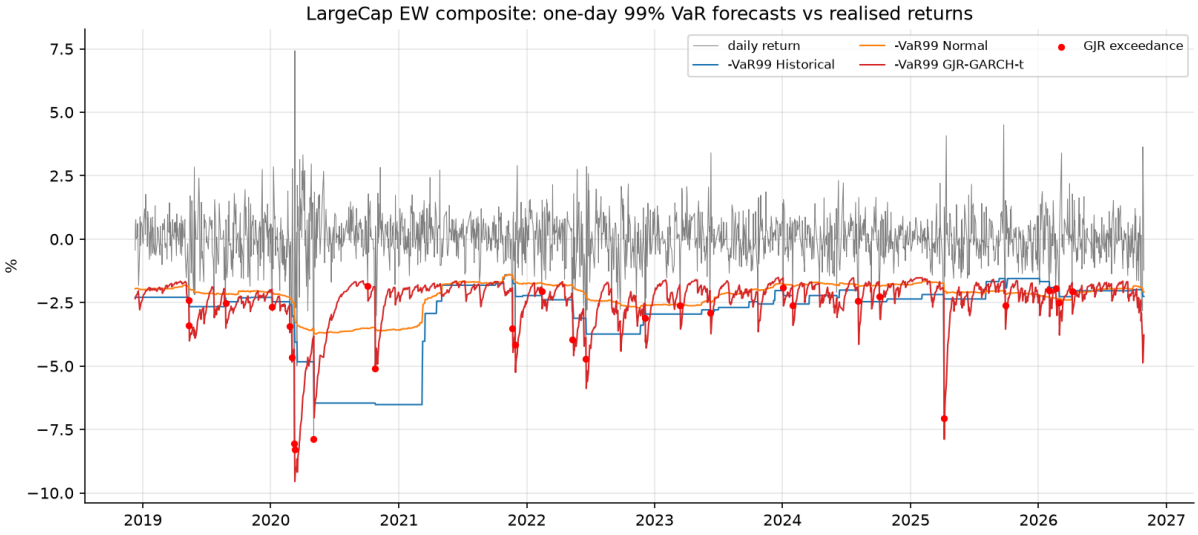


Figure 6. One-day 99% VaR forecasts and exceedances for the LargeCap EW composite.

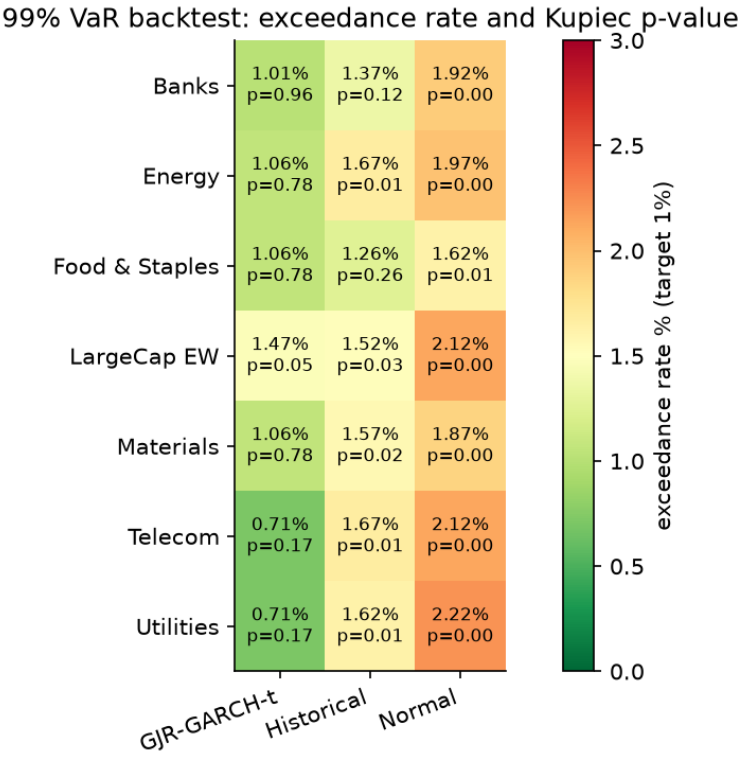


Figure 7. Kupiec POF p-values by series, method and confidence level.

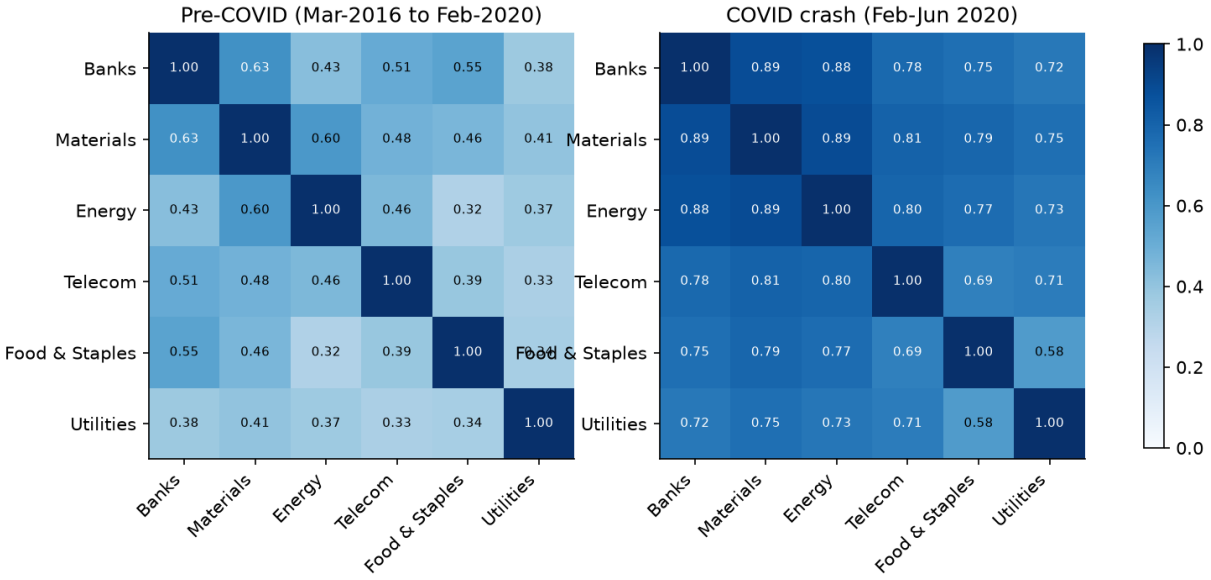


Figure 8. Sector correlation matrices by regime.